Descriptive Statistics Project.

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Objectives:

I shall:

(1.) Dataset description: Describe the dataset.

(2.) Dataset Presentation: Present the dataset using a histogram.

(3.) Measures of Center: Calculate and Interpret the measures of center.

(4.) Measures of Spread: Calculate and Interpret the measures of spread.

(5.) Measures of Position: Calculate and Interpret the measures of position.

Description of Project:

Descriptive Statistics of the monthly maximum precipitation for Ashland Virginia from January 2023- December 2023.

Direct webpage: https://www.weather.gov/wrh/Climate?wfo=akq

<u>Navigation:</u>

Insert these in prompts on webpage:

Location: Ashland, VA

Product: Monthly Summarized Data

Year Range: 2023-2023 Variable: Precipitation Summary: Daily Maximum

GO

<u>Dataset:</u> Precipitation amount in inches

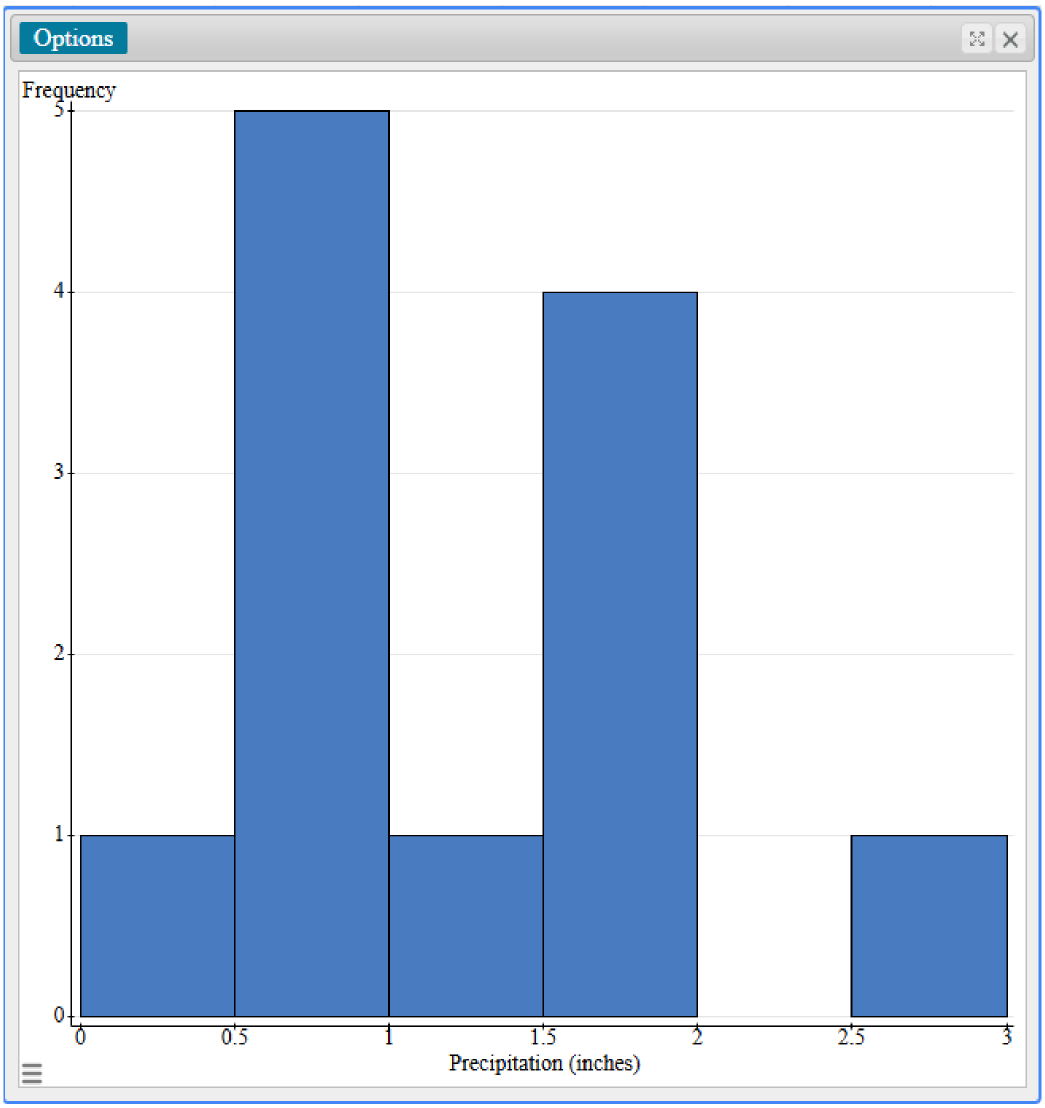
<u>Dataset of monthly maximum precipitation for Ashland, VA in 2023 (in inches):</u>

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2023	0.58	1.13	0.21	1.82	0.61	0.98	1.82	0.95	2.61	0.69	1.52	1.97

0.58, 1.13, 0.21, 1.82, 0.61, 0.98, 1.82, 0.95, 2.61, 0.69, 1.52, 1.97

Data Presentation.

Precipitation (inches)	var2
0.58	January
1.13	Febuary
0.21	March
1.82	April
0.61	May
0.98	June
1.82	July
0.95	August
2.61	September
0.69	October
1.52	November
1.97	December



Notable Information.

[0.5, 1)

5 months have precipitation between 0.5 (included) and 1 inch (excluded).

[1.5, 2)

4 months have precipitation between 1.5 (included) and 2 inches (excluded).

Data Analysis.

Summary statistics:

Column	n	Sum	Mean	Median	Mode	Min	Max	Range	Variance	Std. dev.	Q1	Q3	IQR
Precipitation (inches)	12	14.89	1.2408333	1.055	1.82	0.21	2.61	2.4	0.50188106	0.70843564	0.65	1.82	1.17

We shall use formulas, Pearson StatCrunch software, and Texas Instrument (TI-84CE) calculator.

We shall begin with the measure of center. I shall round all final answers to two decimal places as needed.

Measures of Center

$0.58, 1.13, 0.21, 1.82, 0.61, 0.98, 1.82, 0.95, 2.61, 0.69, 1.52, 1.97$

$Sample\ size = 12$

$$Mean = \frac{Sum\ of\ values}{sample\ size} = \frac{14.84}{12} = 1.24033333 \cong 1.24$$

The average monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is approximately 1.24 inches.

Median: I shall arrange the dataset in ascending order.

$0.21, 0.58, 0.61, 0.69, 0.95, 0.98, 1.13, 1.52, 1.82, 1.82, 1.97, 2.61$

$$Median = \frac{0.98 + 1.13}{2} = 1.055 \cong 1.06$$

The median monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is approximately 1.06 inches.

Mode: I shall determine the most frequent value in the dataset.

$Mode = 1.82$

April and July both had 1.82 inches monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023. The rest had different precipitation values. Hence, the mode is 1.82 inches.

Midrange:

$$Midrange = \frac{Minimum + Maximum}{2} = \frac{0.21 + 2.61}{2} = 1.41$$

The average of the minimum and the maximum monthly precipitation for Ashland, Virginia from January, 2023 to December, 2023 is 1.41 inches.

Measures of Spread.

These are the: Range, Variance, Standard Deviation, and Interquartile Range.

$$Range = Maximum - Minimum$$

$$Range = 2.61 - 0.21 = 2.40$$

The difference between the maximum value and the minimum value of the monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is 2.4 inches.

In calculating the variance and standard deviation, all intermediate values will be rounded to 5 decimal places.

$Precipitation, x$	$x - mean$	$(x - mean)^2$
0.58	-0.66083	0.43670
1.13	-0.11083	0.01228
0.21	-1.03083	1.06262
1.82	0.57917	0.33543
0.61	-0.63083	0.39795
0.98	-0.26083	0.06803
1.82	0.57917	0.33543
0.95	-0.29083	0.08458
2.61	1.36917	1.87462
0.69	-0.55083	0.30342
1.52	0.27917	0.07793
1.97	0.72917	0.53168
$\Sigma(x - Mean)^2 = 5.520691667$		

$$Variance = \frac{\Sigma(x - Mean)^2}{n - 1} = \frac{5.520691667}{12 - 1} = 0.5018810606 \text{ inches}^2$$

$Standard\ Deviation = \sqrt{Variance} = \sqrt{0.5018810606} =$

$0.7084356432\ inches$

There is not much spread in the monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023.

Interquartile Range, IQR.

The dataset arranged in ascending order is:

$0.21, 0.58, 0.61, 0.69, 0.95, 0.98, 1.13, 1.52, 1.82, 1.82, 1.97, 2.61$

$n = 12$

$\frac{1}{4} * 12 = 3$

$3rd\ position = 0.61$

$4\text{th}\ position = 0.69$

$Lower\ Quartile, Q_1 = \frac{0.61 + 0.69}{2} = 0.65\ inches$

$\frac{3}{4} * 12 = 9$

$9\text{th } position = 0.61$

$10\text{th } position = 0.69$

$$Upper\ Quartile, Q_1 = \frac{1.82 + 1.82}{2} = 1.82\ inches$$

$$IQR = Q_3 - Q_1 = 1.82 - 0.65 = 1.17\ inches$$

The variation in 50% of the monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is 1.17 inches.

Measures of Position.

I shall use the five-number summary of data.

$Minimum = 0.21$

The minimum monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is 0.21 inches.

$First\ Quartile, Q_1 = 0.65$

25% of the monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is below 0.65 inches.

$Middle\ Quartile, Q_2 = 1.055$

50% of the monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is below 1.055 inches

$Upper\ Quartile, Q_3 = 1.82$

75% of the monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is below 1.82 inches.

$Maximum = 2.61$

The maximum monthly maximum precipitation for Ashland, Virginia from January, 2023 to December, 2023 is 2.61 inches.

References (MLA 9):

Chukwuemeka, Samuel. "Descriptive Statistics Projects." *Descriptive Statistics*, conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html. Accessed 20 Nov. 2025.

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